

Beyond Collision Classifiers: Control Barrier Functions for Safety-Aware Latent-Space Motion Planning

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by

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Certificate

This is to certify that the thesis titled *Beyond Collision Classifiers: Control Barrier Functions for Safety-Aware Latent-Space Motion Planning* submitted by **Ravula Chandrakiran**, to the Indian Institute of Science, Bengaluru, in partial fulfillment for the award of the degree of **Master of Technology** in **Artificial Intelligence** is a bona fide record of the original work carried out by him/her under my supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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I declare that this thesis titled *Beyond Collision Classifiers: Control Barrier Functions for Safety-Aware Latent-Space Motion Planning* submitted in partial fulfillment for the award of the degree of **Master of Technology in Artificial Intelligence** is a record of the original work carried out by me under the supervision of **Prof. Ravi Prakash**, and has not formed the basis for the award of any degree, diploma, associateship, fellowship, or other titles in this or any other Institution or University of higher learning. In keeping with the ethical practice in reporting scientific information, due acknowledgments have been made wherever the findings of others have been cited.

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Abstract

Motion planning for high-dimensional robotic manipulators is a fundamental challenge in robotics. Classical sampling-based planners struggle with the curse of dimensionality, while optimization-based methods are susceptible to local minima and require expensive geometric collision checking. Learned latent-space representations offer a promising alternative by compressing the configuration space into a compact, differentiable manifold, enabling fast gradient-based planning without explicit inverse kinematics.

Existing latent-space path planning (LSPP) frameworks rely on binary collision classifiers to enforce obstacle avoidance. However, classifier-based safety signals provide weak or saturated gradients near obstacle boundaries and offer no principled temporal structure or forward-invariance guarantees. This often leads to optimization stalling and premature convergence in cluttered environments. Furthermore, the integration of formal control-theoretic safety bounds into learned generative spaces remains underexplored.

This thesis reproduces a state-of-the-art LSPP baseline for a 7-DoF Franka Panda manipulator and extends it by replacing the binary collision classifier with a learned neural Control Barrier Function (CBF). We formulate a novel three-term training objective to learn the barrier function and introduce a closed-form, first-order gradient correction mechanism that explicitly enforces forward invariance during the latent-space Activation Maximization planning phase. The proposed framework is evaluated across three variants of increasing generality: a fixed-obstacle barrier, a position-conditioned barrier, and a general random-obstacle barrier.

Empirical results demonstrate that the CBF formulation achieves a 75.6% planning success rate on the fixed-obstacle task—outperforming the best-tuned collision classifier by 12.4 percentage points—and an 83.8% success rate on the multi-position task under identical planning conditions. However, performance degradation in the fully general random-obstacle setting reveals critical insights regarding training distribution mismatch and topological complexity in high-dimensional conditional barrier networks.

This work establishes the feasibility of CBF-based safety constraints in learned latent spaces, provides a systematic characterization of the safety-performance tradeoff, and lays the theoretical and empirical groundwork for future closed-loop, model-predictive formulations in latent-space manipulation planning.

Contents

Certificate	i
Declaration	ii
Acknowledgements	iii
Abstract	iv
Table of Contents	v
List of Figures	viii
List of Tables	ix
List of Algorithms	x
Abbreviations	xi
1 Introduction	1
1.1 Latent-Space Planning as an Alternative	1
1.2 Limitations of Classifier-Based Safety Constraints	2
1.3 Control Barrier Functions as a Safety Framework	2
1.4 Thesis Contributions	2
1.5 Thesis Organization	3
2 Background and Related Work	4
2.1 Motion Planning for Robotic Manipulators	4
2.2 Variational Autoencoders for Representation Learning	4
2.3 Latent-Space Path Planning	5

2.4	Control Barrier Functions	5
3	Latent-Space Path Planning — Baseline Framework	7
3.1	Robot State Representation	7
3.2	VAE Architecture and Training	7
3.3	Collision Classifier Architecture	8
3.4	Activation Maximization Planning	8
3.5	Two-Stage Evaluation Pipeline	9
3.6	Baseline Experimental Results	9
4	Control Barrier Functions in Latent Space	12
4.1	Motivation for CBFs in Latent Space	12
4.2	CBF Training Data Generation	12
4.3	Neural CBF Architecture and Training	14
4.4	CBF Safety Correction at Inference	14
4.5	Analysis of the α Parameter	16
5	Evaluation Framework and Experimental Setup	17
5.1	Experimental Objectives	17
5.2	Dataset and Environment	17
5.3	Experimental Settings	18
5.4	Ablation Design and Metrics	18
6	Experimental Results and Analysis	19
6.1	Fixed-Obstacle CBF vs. Classifier	19
6.2	Multi-Position Generalization Results	20
6.3	General Obstacle Planning Results	21
6.4	Computational Analysis	21
7	Conclusion and Future Work	23
7.1	Summary of Contributions	23
7.2	Limitations	23
7.3	Future Work	24
	Bibliography	24
	Appendices	27

A	Implementation and Hyperparameter Details	27
A.1	A.1 Variational Autoencoder (VAE)	27
A.2	A.2 Collision Classifier (VAEObstacleBCE)	28
A.3	A.3 Control Barrier Function Networks	28
B	Hyperparameter Configurations	30
B.1	B.1 Training Configurations	30
B.2	B.2 Planning and Inference Configurations	31
C	Dataset and Environment Properties	32
C.1	C.1 Free-Space and Collision Datasets	32
C.2	C.2 CBF State and Transition Datasets	32
C.3	C.3 Simulation and Evaluation Geometry	33

List of Figures

6.1	Success, Goal, and CF Rate vs α	19
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List of Tables

3.1	Baseline Experimental Results	10
6.1	Key α Ablation Results for FixedBarrierNet	19
6.2	Best CBF vs. Best Classifier	20
6.3	Key α Ablation Results for MultiPosBarrierNet	20
6.4	Per-Position Breakdown at Optimal $\alpha = 0.10$	21
6.5	Key α Ablation Results for General BarrierNet	21
B.1	Model Training Hyperparameters	30
B.2	Activation Maximization Planning Hyperparameters	31

List of Algorithms

3.1	Latent-Space Activation Maximization (AM) for Path Planning	9
3.2	Two-Stage Evaluation and Validation Pipeline	10
4.1	Dual-Dataset Generation for Spatial and Temporal CBF Training	13
4.2	Closed-Form CBF Safety Correction via Predictor-Corrector Optimization	15

Abbreviations

AM	Activation Maximization
AUC	Area Under Curve
BCE	Binary Cross-Entropy
CBF	Control Barrier Function
CF	Collision-Free
CHOMP	Covariant Hamiltonian Optimization for Motion Planning
DoF	Degree-of-Freedom
ELBO	Evidence Lower Bound
ELU	Exponential Linear Unit
GECO	Generalized ELBO with Constrained Optimization
KL	Kullback-Leibler (Divergence)
LSPP	Latent-Space Path Planning
MPC	Model Predictive Control
MSE	Mean Squared Error
PRM	Probabilistic Roadmaps
RRT	Rapidly-exploring Random Trees
VAE	Variational Autoencoder

Chapter 1

Introduction

Motion planning for robotic manipulators operating in high-dimensional configuration spaces remains a formidable computational challenge. A 7-Degree-of-Freedom (DoF) manipulator, such as the Franka Emika Panda, must compute collision-free trajectories from a start configuration to a goal while strictly satisfying joint limits, avoiding workspace obstacles, and remaining kinematically feasible. Because the configuration space is high-dimensional and the mapping from joint space to task space is nonlinear and non-bijective, exhaustive search is intractable. While classical sampling-based methods like Rapidly-exploring Random Trees (RRT) and Probabilistic Roadmaps (PRM) offer probabilistic completeness, their runtime scales poorly with dimensionality. Optimization-based planners such as Covariant Hamiltonian Optimization for Motion Planning (CHOMP) and TrajOpt alleviate some scalability issues, but they are highly sensitive to initialization, prone to local minima, and rely heavily on computationally expensive distance-field queries or exact geometric collision checking at every iteration.

1.1 Latent-Space Planning as an Alternative

Learned latent representations offer a compelling alternative by compressing the planning problem into a lower-dimensional, structurally smooth manifold. A Variational Autoencoder (VAE), trained on a dataset of valid robot configurations, can learn a compact latent space that implicitly captures kinematic feasibility and the multimodality of inverse kinematics. Gradient-based optimization—specifically, Activation Maximization (AM)—can then be performed directly in this continuous, differentiable space to simultaneously optimize goal-reaching and obstacle-avoidance objectives.

This thesis builds directly upon the foundational framework proposed by Hung et al. (2022) in “Reaching Through Latent Space: From Joint Statistics to Path Planning in Manipulation.” The reference architecture utilizes a VAE to encode both joint configurations and end-effector positions, combined with a learned binary collision classifier. Planning is executed by steering

the latent state toward regions that minimize distance to the goal while maximizing the classifier’s predicted safety. A complete reproduction of this baseline framework was achieved, yielding a 70.5% success rate in general planning scenarios.

1.2 Limitations of Classifier-Based Safety Constraints

Despite the computational efficiency of the baseline LSPP framework, relying on a binary collision classifier to enforce obstacle avoidance introduces significant structural weaknesses. First, the sigmoid output of a classifier provides a poor gradient landscape: far from an obstacle, the gradient is practically zero, offering no anticipatory guidance; inside an obstacle, the gradient violently spikes. This “cliff-edge” profile routinely traps the trajectory optimizer in local minima. Second, classifiers lack temporal structure. They evaluate static configurations without considering the trajectory’s flow, providing no guarantee that a state deemed “safe” at step k will not transition into a catastrophic collision at step $k + 1$.

1.3 Control Barrier Functions as a Safety Framework

To address the fundamental limitations of classifier-guided planning, this thesis proposes the integration of Control Barrier Functions (CBFs). A Control Barrier Function, $B(z)$, is a continuous real-valued function where $B \geq 0$ denotes a safe state. Crucially, CBFs mandate a decrease condition or forward invariance: if the system is currently safe, future states must remain safe at a controlled decay rate, governed by a parameter α . By mapping the CBF directly into the learned latent space, we transform a weak, soft collision penalty into a hard, closed-form safety correction step. This mechanism actively filters nominal goal-directed steps, projecting them safely away from learned obstacle boundaries without interfering with the primary optimization objective.

1.4 Thesis Contributions

The primary contributions of this thesis are as follows:

- A complete reproduction of the latent-space path planning baseline for a 7-DoF Franka Panda manipulator, featuring a two-stage evaluation pipeline backed by exact geometric collision verification.
- The formulation of a neural CBF framework operating entirely within a learned latent space,

governed by a novel three-term training objective (safe sign, unsafe sign, and decrease condition).

- The derivation and implementation of a closed-form, first-order gradient correction mechanism for safe latent trajectory generation.
- The design of three experimental CBF variants of increasing generality: a fixed-obstacle barrier $B(z)$, a position-conditioned barrier $B(z, x, y)$, and a general obstacle-conditioned barrier $B(z, o)$.
- A systematic empirical characterization of the α safety-performance tradeoff, demonstrating that the proposed CBF formulation outperforms the best-tuned collision classifier by 12.4 percentage points on fixed-obstacle tasks and achieves 83.8% success on multi-position tasks.

1.5 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews the theoretical background of VAEs, latent-space planning, and CBFs. Chapter 3 details the architecture, training, and performance of the reproduced baseline framework. Chapter 4 introduces the proposed neural CBF formulation, the dual-dataset generation procedure, and the mathematics of the inference-time safety correction. Chapter 5 outlines the experimental setup, evaluation metrics, and ablation methodology. Chapter 6 presents a comprehensive analysis of the empirical results across all obstacle variants. Finally, Chapter 7 concludes the thesis with a discussion of current limitations and future research directions.

Chapter 2

Background and Related Work

This chapter reviews the theoretical foundations and related literature necessary to contextualize the proposed approach. The discussion is organized into three domains: motion planning for manipulators, generative representation learning, and safe control via barrier functions.

2.1 Motion Planning for Robotic Manipulators

Formally, the motion planning problem requires finding a continuous trajectory $\tau(t) \in \mathcal{C}_{free}$ from an initial configuration q_{start} to a target q_{goal} , where $\mathcal{C}_{free}$ is the collision-free subset of the configuration space.

Sampling-based algorithms, notably RRT and PRM, bypass explicit configuration space construction by randomly sampling configurations and connecting them via local planners. While probabilistically complete, the volume of high-dimensional configuration spaces results in highly variable planning times and suboptimal, jerky trajectories requiring post-processing. Trajectory optimization approaches, conversely, formulate planning as the minimization of a cost functional combining path smoothness and obstacle avoidance. However, representing complex geometric constraints analytically is difficult, and non-convex environments frequently trap the optimizer in local minima. Both paradigms suffer from a heavy reliance on continuous forward kinematics and geometric intersection tests during the planning loop.

2.2 Variational Autoencoders for Representation Learning

To bypass explicit inverse kinematics and dimensional scaling issues, generative models—specifically Variational Autoencoders (VAEs)—can be employed to learn a compressed representation of the state space. A standard VAE optimizes the Evidence Lower Bound (ELBO), balancing reconstruction accuracy with a Kullback-Leibler (KL) divergence penalty that regularizes the latent space z

toward a prior $p(z) \sim \mathcal{N}(0, I)$.

In the context of physical state representation, the standard ELBO often fails to guarantee strict reconstruction fidelity, which is paramount for robot kinematics. Therefore, this work utilizes the Generalized Endogenous Control (GECO) framework. GECO treats reconstruction quality as a constrained optimization problem rather than a fixed-weight ELBO term. The GECO Lagrangian is formulated as:

$$\mathcal{L}_{GECO} = -D_{KL}[q_\phi(z|x)||p(z)] + \lambda \mathbb{E}_z[\max(0, \text{MSE}(x, \hat{x}) - g_{goal})] \quad (2.1)$$

where λ acts as an adaptive Lagrange multiplier updated via an exponential moving average. If the mean squared error (MSE) exceeds a strict physical threshold g_{goal} , λ increases, penalizing poor reconstruction. If the constraint is met, λ decreases, allowing the network to prioritize structural compression.

2.3 Latent-Space Path Planning

Latent-Space Path Planning (LSPP) shifts the trajectory optimization problem entirely into the learned variable z . As introduced by Hung et al. (2022), the robot state is parameterized as $x = (q, e)$, comprising both joint angles and end-effector coordinates. Given a start configuration q_{start} and a target position e_{target} , the planner solves:

$$\min_z (\|\hat{e}(z) - e_{target}\|^2 + \lambda_{prior}\mathcal{L}_{prior} + \lambda_{obs}\mathcal{L}_{obs}) \quad (2.2)$$

This procedure, termed Activation Maximization (AM), leverages the VAE’s differentiability. A binary collision classifier provides $\mathcal{L}_{obs}$, steering z away from states predicted to cause collisions. While elegant, the reliance on a classifier inherently limits the gradient quality available to the optimizer.

2.4 Control Barrier Functions

Control Barrier Functions provide a mathematically rigorous alternative for safety-critical control. For a nonlinear continuous dynamical system $\dot{x} = f(x, u)$, a safe set $\mathcal{C}$ is defined as the superlevel set of a continuously differentiable function $B(x)$: $\mathcal{C} = \{x \in \mathbb{R}^n : B(x) \geq 0\}$.

The set $\mathcal{C}$ is rendered forward-invariant if there exists an extended class- $\mathcal{K}$ function α such that:

$$\dot{B}(x) + \alpha(B(x)) \geq 0 \quad (2.3)$$

For discrete-time trajectory generation, this condition is adapted to ensure the barrier function does not decay faster than a prescribed rate. Given a timestep Δt , the discrete forward invariance condition requires:

$$B(z_{k+1}) \geq (1 - \alpha\Delta t)B(z_k) \quad (2.4)$$

While CBFs have been extensively studied in control theory, their application directly within the latent space of generative neural networks represents a novel intersection of deep learning and safe control theory, serving as the core contribution of this thesis.

Chapter 3

Latent-Space Path Planning — Baseline Framework

This chapter details the architecture, training procedure, and baseline performance of the standard LSPP framework. The reproduction serves as the benchmark against which the proposed Control Barrier Function extensions are subsequently evaluated.

3.1 Robot State Representation

The framework models the state of the 7-DoF Franka Panda manipulator as a concatenated vector $x = (q, e) \in \mathbb{R}^{10}$, where $q \in \mathbb{R}^7$ represents the joint configuration in radians, and $e \in \mathbb{R}^3$ represents the Cartesian end-effector position obtained via forward kinematics. Incorporating e directly into the state eliminates the need for the planner to solve complex inverse kinematics; the spatial relationship between q and e is implicitly learned by the VAE.

To stabilize training, all ten dimensions are individually standardized (zero-mean, unit-variance) using statistics computed from the training dataset. It is critical to note that cross-dataset normalization is enforced: when training the collision classifier, the state variables are normalized using the free-space dataset’s statistics to ensure the classifier operates in the exact mathematical space as the VAE.

3.2 VAE Architecture and Training

The VAE maps the physical state $x \in \mathbb{R}^{10}$ to a compact latent representation $z \in \mathbb{R}^7$.

The encoder comprises four hidden layers of 2048 units each, utilizing Exponential Linear Unit (ELU) activations. It outputs the parameters of a multivariate Gaussian: $\mu \in \mathbb{R}^7$ and $\log \sigma^2 \in \mathbb{R}^7$. Reparameterization is applied as $z = \mu + \epsilon \cdot \sigma$, where $\epsilon \sim \mathcal{N}(0, I)$. To ensure numerical

stability, variance is constrained via $\sigma^2 = \text{softplus}(\text{fc}_{22}(h)) + 1e^{-5}$. The decoder mirrors the encoder structure, mapping z back to $\hat{x} = (\hat{q}, \hat{e})$.

The model was trained for 16,000 epochs using the Adam optimizer (learning rate $1e^{-4}$ and batch size 4096). The GECO threshold was strictly set to $g_{goal} = 0.0008$. Checkpoint selection was determined not by minimum validation loss, but by an empirical Activation Maximization Area Under Curve (AUC) metric, guaranteeing the selected model inherently supported stable gradient descent planning.

3.3 Collision Classifier Architecture

Obstacle avoidance in the baseline is handled by VAEObstacleBCE, a neural classifier appended to the frozen VAE encoder. The classifier head accepts an 11-dimensional input vector: the 7D latent code z concatenated with a 4D normalized obstacle descriptor $o = [x, y, h, r]$ (representing the planar coordinates, height, and radius of a cylindrical obstacle).

The classifier consists of four hidden layers of 2048 units and outputs a single logit, which is passed through a sigmoid function to yield $P(\text{collision} \mid z, o)$. The network is trained using standard Binary Cross-Entropy (BCE) over 16,000 epochs.

3.4 Activation Maximization Planning

Given an initial joint configuration q_{start} and a target end-effector position e_{target} , AM performs gradient descent on z . The objective function optimized at each step t is:

$$\mathcal{L}_{AM} = \|\hat{e}(z) - e_{target}\|^2 + \lambda_{prior}(0.5\|z\|^2) - \lambda_{obs} \sum_i \log(1 - \sigma(\text{logit}(z, o_i)/T)) \quad (3.1)$$

The first term drives the end-effector toward the target. The second term regularizes z toward the VAE prior to prevent the generation of out-of-distribution, kinematically invalid states. The third term minimizes collision probability, scaled by temperature $T = 3.0$ to soften the classifier's gradients.

A critical design choice involves initializing the optimizer with a sampled latent vector z rather than the deterministic mean μ . This implicitly provides necessary exploration noise in the highly non-convex loss landscape. The optimization is executed for a maximum of 300 steps using Adam with a learning rate of 0.15.

Algorithm 3.1: Latent-Space Activation Maximization (AM) for Path Planning

Input: Initial configuration q_{start} , target e_{target} , obstacles $\mathcal{O}$, max steps $N_{max} = 300$, learning rate $\eta = 0.15$

Output: Path $\mathcal{P}$

```
1  $z_{init} \sim \text{Encoder}(q_{start});$  // Sample from learned latent distribution
2 Initialize  $z \leftarrow z_{init};$ 
3 Initialize Adam optimizer for  $z$  with learning rate  $\eta$ ;
4 Path  $\mathcal{P} \leftarrow \emptyset;$ 
5 for  $t = 1$  to  $N_{max}$  do
6   Compute  $\mathcal{L}_{AM}$  using Equation 3.1;
7    $\nabla z \leftarrow$  Compute gradient of  $\mathcal{L}_{AM}$  w.r.t  $z$ ;
8    $z \leftarrow z - \text{AdamUpdate}(\nabla z);$ 
9    $(\hat{q}_t, \hat{e}_t) \leftarrow \text{Decoder}(z);$ 
10   $\mathcal{P} \leftarrow \mathcal{P} \cup \{(\hat{q}_t, \hat{e}_t)\};$ 
11  if  $\|\hat{e}_t - e_{target}\|_2 < \text{threshold}$  then
12    break;
13  end
14 end
15 return Path  $\mathcal{P};$  // Goal convergence achieved
```

3.5 Two-Stage Evaluation Pipeline

To ensure rigorous performance reporting, a strict two-stage methodology is employed. During Stage 1 (Planning), the continuous gradient from the neural classifier steers the latent trajectory. During Stage 2 (Validation), the resulting joint trajectory is decoded and evaluated against an independent, ground-truth geometric simulator (Robo3D). Robo3D utilizes capsule-based collision checking to compute exact minimum distances between the robot links and obstacles. A trajectory is deemed successful only if the target is reached within a 10mm threshold and no waypoint intersects an obstacle in the geometric simulation.

3.6 Baseline Experimental Results

The baseline framework was evaluated across 1,000 randomly generated scenarios. As summarized in Table 3.1, the system achieved a Success Rate of 70.5%.

Failure mode analysis reveals a 9.3% non-convergence gap (the optimizer stalls and fails to reach the goal entirely) and an 8.7% safety failure rate (paths reached the goal but clipped an obstacle). The primary bottleneck is the gradient quality near obstacle boundaries. The classifier's

Algorithm 3.2: Two-Stage Evaluation and Validation Pipeline

Input: Set of test scenarios $\mathcal{S}$, trained VAE, trained Classifier, geometric simulator (Robo3D)

Output: Success Rate

```
1  $SuccessCount \leftarrow 0$ ;  
2 foreach scenario  $s \in \mathcal{S}$  do  
3   Stage 1: Differentiable Planning;  
4    $\mathcal{P} \leftarrow$  Run Algorithm 3.1 for scenario  $s$ ;  
5   Stage 2: Geometric Validation;  
6    $CollisionFree \leftarrow \text{True}$ ;  
7   foreach waypoint  $(\hat{q}, \hat{e}) \in \mathcal{P}$  do  
8     if  $Robo3D.check\_collision(\hat{q}, s.obstacles)$  then  
9        $CollisionFree \leftarrow \text{False}$ ;  
10      break;  
11    end  
12  end  
13  if  $CollisionFree$  and  $\|\mathcal{P}_{final}.\hat{e} - s.e_{target}\|_2 < threshold$  then  
14     $SuccessCount \leftarrow SuccessCount + 1$ ;  
15  end  
16 end  
17 return  $SuccessRate = SuccessCount / |\mathcal{S}|$ ;
```

Table 3.1: Baseline Experimental Results

Success Rate	Goal Reached	Collision-Free	Avg Time (ms)
70.5%	79.2%	79.8%	186.3

gradients are either imperceptible far from the obstacle or overwhelmingly dominant upon entry, trapping the optimizer in local minima. This structural limitation necessitates the formulation of a more robust safety mechanism.

Chapter 4

Control Barrier Functions in Latent Space

This chapter details the theoretical formulation and implementation of the proposed neural Control Barrier Function in latent space. By abandoning the binary BCE objective, we construct a continuous, structurally aware safety boundary that guarantees forward invariance during planning.

4.1 Motivation for CBFs in Latent Space

Replacing the collision classifier with a CBF resolves three fundamental issues:

- **Gradient Quality:** The BCE loss term saturates at high and low confidence intervals. A CBF, engineered as an unbounded signed distance function, provides dense, non-vanishing gradient vectors $\nabla_z B(z, o)$ that precisely reflect local boundary geometry.
- **Temporal Structure:** Classifiers operate entirely independently of time. The discrete CBF condition mathematically couples z_k to z_{k+1} , explicitly prohibiting latent steps that catastrophically diminish the safety margin in a single iteration.
- **Constraint Modularity:** In the baseline, goal seeking and safety compete directly within a single scalar loss function. The CBF decouples this: a nominal goal-directed step is computed freely, and safety is applied post-hoc as a mathematical projection, preventing gradient interference.

4.2 CBF Training Data Generation

Training a barrier function requires resolving both spatial geometry (where is the obstacle?) and temporal dynamics (how does the planner move?). Consequently, a dual-dataset approach was developed.

- **State-Label Dataset:** Represents the spatial boundary. Collision configurations are passed deterministically through the VAE to generate pairs of (z, o, label) . To sharpen the decision boundary, the dataset is augmented with samples explicitly generated near the obstacle surfaces.
- **Transition Dataset:** Represents the temporal dynamics. Critically, transitions (z_k, z_{nom}) are generated by running a nominal planner (driven only by the goal and prior loss, ignoring obstacles). This design choice is vital: the CBF must learn to enforce the decrease condition specifically on the behavior of the naive goal-seeking optimizer it will correct at inference time.

Algorithm 4.1: Dual-Dataset Generation for Spatial and Temporal CBF Training

Input: Base collision dataset $\mathcal{D}_{base}$, Nominal Planner, Geometric Simulator

Output: $\mathcal{D}_{state}, \mathcal{D}_{trans}$

```

1 Part 1: State-Label Dataset Generation;
2  $\mathcal{D}_{state} \leftarrow \emptyset$ ;
3 foreach physical configuration  $(q, o, \text{label}) \in \mathcal{D}_{base}$  do
4    $\mu, \sigma \leftarrow \text{Encoder}(q)$ ;
5    $\mathcal{D}_{state} \leftarrow \mathcal{D}_{state} \cup \{(\mu, o, \text{label})\}$ ;
6   ; // Deterministic encoding
7 end
8 Part 2: Transition Dataset Generation;
9  $\mathcal{D}_{trans} \leftarrow \emptyset$ ;
10 for scenario  $i = 1$  to  $N_{scenarios}$  do
11   Initialize  $z_0 \sim \text{Encoder}(q_{start})$ ;
12   for step  $k = 0$  to  $K_{max}$  do
13      $z_{nom} \leftarrow z_k - \text{AdamUpdate}(\nabla(\mathcal{L}_{goal} + \mathcal{L}_{prior}))$ ;
14     ; // Obstacle-blind step
15      $\text{label}_k \leftarrow \text{Simulator.check\_safety}(z_k)$ ;
16      $\text{label}_{nom} \leftarrow \text{Simulator.check\_safety}(z_{nom})$ ;
17      $\mathcal{D}_{trans} \leftarrow \mathcal{D}_{trans} \cup \{(z_k, z_{nom}, o, \text{label}_k, \text{label}_{nom})\}$ ;
18      $z_{k+1} \leftarrow z_{nom}$ ;
19   end
20 end
21 return  $\mathcal{D}_{state}, \mathcal{D}_{trans}$ ;

```

4.3 Neural CBF Architecture and Training

The general barrier function BarrierNet takes an 11D input (7D latent vector z , 4D obstacle o) and outputs a continuous scalar $B \in (-\infty, \infty)$ through four layers of 2048 ELU units. Unlike the classifier, no final sigmoid activation is applied. The sign convention dictates that $B \geq 0$ represents the safe set.

The network is trained to minimize a novel three-term loss function:

$$\mathcal{L}_{total} = \lambda_s \mathbb{E}_{safe}[\text{ReLU}(-B + \gamma)] + \lambda_u \mathbb{E}_{unsafe}[\text{ReLU}(B + \gamma)] + \lambda_d \mathbb{E}_{trans}[\text{ReLU}(B_{target} - B_{nom})] \quad (4.1)$$

Where $B_{target} = (1 - \alpha \Delta t)B(z_k)$.

A symmetric margin $\gamma = 1.0$ is enforced for the spatial losses, demanding that safe states satisfy $B \geq 1$ and unsafe states $B \leq -1$. This 2.0-unit decision gap creates a mathematical buffer, providing robustness against VAE decoding variance.

To isolate the difficulty of obstacle generalization, three variants were trained:

- **FixedBarrierNet** $B(z)$: 7D input, evaluating safety against a static, globally fixed obstacle.
- **MultiPosBarrierNet** $B(z, x, y)$: 9D input, conditioning only on the planar coordinates of the obstacle.
- **BarrierNet** $B(z, o)$: 11D input, conditioning on full random configurations.

4.4 CBF Safety Correction at Inference

During planning, the $\mathcal{L}_{obs}$ penalty is completely removed. Instead, a predictor-corrector mechanism is utilized. At step k , a nominal step is taken: $z_{nom} \leftarrow \text{Adam}(z_k, \nabla(\mathcal{L}_{goal} + \mathcal{L}_{prior}))$.

The CBF constraint evaluates if $B(z_{nom}) \geq B_{target}$. If violated, a safe state is projected using the barrier gradient:

$$z_{safe} = z_{nom} + \lambda \nabla_z B(z_{nom}, o) \quad (4.2)$$

Assuming a first-order Taylor approximation around z_{nom} :

$$B(z_{nom} + \lambda \nabla B) \approx B(z_{nom}) + \lambda \|\nabla B(z_{nom})\|^2 \quad (4.3)$$

Setting this equal to B_{target} yields the minimum-norm correction multiplier:

$$\lambda = \max \left(0, \frac{B_{target} - B(z_{nom})}{\|\nabla_z B(z_{nom}, o)\|^2 + \epsilon} \right) \quad (4.4)$$

Adam State Reset: A critical engineering implementation is required to stabilize this loop. When the safety correction modifies `z.data`, the internal momentum variables maintained by the Adam optimizer become misaligned with the new physical coordinate. To prevent accumulated momentum from violently fighting the safety projection in subsequent iterations, the Adam optimizer’s internal state for z is flushed (`optimizer.state[z] = {}`) whenever $\lambda > 0$.

Algorithm 4.2: Closed-Form CBF Safety Correction via Predictor-Corrector Optimization

Input: Current latent state z_k , obstacle o , decay rate α , timestep Δt , trained BarrierNet B

Output: Safe latent state z_{safe}

```

1 Predictor Step (Nominal Goal Pursuit);
2  $z_{nom} \leftarrow z_k - \text{AdamUpdate}(\nabla(\mathcal{L}_{goal} + \mathcal{L}_{prior}));$ 
3 Corrector Step (Safety Filter);
4  $B_k \leftarrow B(z_k, o);$ 
5  $B_{target} \leftarrow (1 - \alpha\Delta t)B_k;$ 
6 if  $B(z_{nom}, o) \geq B_{target}$  then
7    $z_{safe} \leftarrow z_{nom};$ 
8 else
9   Compute gradient  $\nabla_z B(z_{nom}, o)$  via autograd;
10   $\lambda \leftarrow \max \left( 0, \frac{B_{target} - B(z_{nom}, o)}{\|\nabla_z B(z_{nom}, o)\|_2^2 + \epsilon} \right);$ 
11   $\lambda \leftarrow \min(\lambda, 1.0);$ 
12   $z_{safe} \leftarrow z_{nom} + \lambda \nabla_z B(z_{nom}, o);$ 
13  optimizer.state[z]  $\leftarrow \emptyset$ ;
14 end
15 return  $z_{safe};$ 

```

4.5 Analysis of the α Parameter

The parameter α fundamentally dictates the safety-performance tradeoff. As $\alpha \rightarrow 0$, the constraint requires $B(z_{k+1}) \geq B(z_k)$. This forbids any decrease in the barrier function, creating an ultra-conservative planner that rapidly stalls in the presence of an obstacle. Conversely, as $\alpha \rightarrow 1$ (assuming $\Delta t = 1$), the constraint only requires $B(z_{nom}) \geq 0$. This permits massive leaps through latent space, routinely violating the local linearity assumption of the Taylor approximation and resulting in geometric collisions upon decoding. Optimal performance therefore lies in an intermediate regime.

Chapter 5

Evaluation Framework and Experimental Setup

This chapter establishes the experimental parameters, datasets, and ablation methodology designed to rigorously evaluate the CBF framework against the classifier baseline.

5.1 Experimental Objectives

The evaluation is structured to answer three precise questions:

- Does the closed-form CBF correction outperform a tuned binary classifier under identical planning conditions?
- How does the decay parameter α quantitatively mediate the safety-performance tradeoff?
- To what extent does the topological complexity of conditioning the barrier network on variable obstacle parameters affect generalizability?

5.2 Dataset and Environment

The workspace is constrained to the physical limitations of the Franka Panda (e.g., $q_1 \in [-166^\circ, 166^\circ]$). The training corpus comprises 100,000 free-space transitions and 100,000 obstacle collision states, uniformly sampled within the workspace bounding volume $x \in [0.4, 0.8]$ m, $y \in [-0.5, 0.5]$ m. The physical validation utilizes Robo3D, simulating exact link capsules and evaluating intersections against cylindrical geometry.

5.3 Experimental Settings

- **Setting 1: Fixed-Obstacle Control.** Designed to isolate the efficacy of the safety mechanism devoid of generalization noise. The obstacle is fixed at $o = [0.5, 0.0, 0.75, 0.1]$ and the target at $e = [0.399, 0.218, 0.656]$. The FixedBarrierNet is compared directly against a FixedCollisionClassifier identical in architectural depth. Planning parameters are stripped of GECO influence (learning rate 0.03, $\lambda_{prior} = 0.01$) to ensure the comparison strictly isolates the safety mechanisms.
- **Setting 2: Multi-Position Generalization.** Evaluates the capacity of MultiPosBarrierNet to interpolate spatial boundaries across four distinct planar coordinates: $[0.4, -0.2]$, $[0.5, 0.0]$, $[0.6, 0.2]$, $[0.5, 0.0]$.
- **Setting 3: General Planning.** Evaluates BarrierNet on 500 randomized scenarios, matching the conditions of the original baseline experiment, to test ultimate generalizability.

5.4 Ablation Design and Metrics

A dense parameter sweep was executed. For all CBF variants, α was swept from 0.00 to 1.00 in precise increments of 0.05 (yielding 21 specific test conditions per variant). For the classifier baseline, a 5×7 grid search over λ_{prior} and $\lambda_{collision}$ was conducted to ensure the CBF was compared against the absolute optimal configuration of the competing method.

Performance is primarily quantified by three metrics computed over 500 scenarios:

- **Success Rate:** The end-effector reaches within 10mm of the target AND zero collisions are detected.
- **Goal Reached Rate:** End-effector converges to the target (ignoring collisions).
- **Collision-Free (CF) Rate:** Zero collisions detected (ignoring convergence).

Chapter 6

Experimental Results and Analysis

This chapter analyzes the empirical data drawn from over 30,000 executed planning scenarios, comparing the CBF architecture against the classifier baseline.

6.1 Fixed-Obstacle CBF vs. Classifier

The efficacy of the pure CBF mechanism is best observed in the fixed-obstacle ablation, removing the confounding variable of obstacle conditioning.

Table 6.1: Key α Ablation Results for FixedBarrierNet

α	Success Rate	Goal Rate	CF Rate	Interv. Rate	Time (ms)
0.00	46.8%	56.4%	88.2%	90.8%	180.8
0.05	74.4%	86.2%	87.0%	67.0%	117.4
0.10	74.4%	88.2%	85.0%	52.3%	99.6
0.15	75.0%	90.0%	83.2%	43.5%	95.6
0.20	75.6%	91.4%	82.0%	37.9%	94.8
0.25	74.0%	92.2%	79.2%	34.1%	92.3
0.30	71.0%	92.2%	75.8%	31.7%	86.8
1.00	58.0%	99.0%	58.4%	19.7%	117.0

Figure 6.1: Success, Goal, and CF Rate vs α

The data reveals a non-monotonic success curve. At the ultra-conservative setting of $\alpha = 0.00$, the intervention rate strikes 90.8%, practically freezing the optimizer. Goal completion collapses to 56.4%, resulting in a low success rate of 46.8%. As α increases, goal completion climbs steadily, reaching 99.0% at $\alpha = 1.00$. However, CF rate decays proportionally due to linear approximation failure on aggressive steps.

The optimal nexus occurs at $\alpha = 0.20$, achieving a 75.6% Success Rate (91.4% Goal, 82.0% CF). At this setting, the intervention rate is a balanced 37.9%.

Table 6.2: Best CBF vs. Best Classifier

Method	Success Rate	Goal Rate	CF Rate	Time (ms)
CBF $\alpha = 0.20$ (Best)	75.6%	91.4%	82.0%	94.8
CBF $\alpha = 0.15$	75.0%	90.0%	83.2%	95.6
CBF $\alpha = 0.10$	74.4%	88.2%	85.0%	99.6
Classifier $\lambda_c = 10.0$ (Best)	63.2%	79.2%	67.2%	110.1
Classifier $\lambda_c = 5.0$	61.2%	78.8%	65.8%	92.9
Classifier $\lambda_c = 0.5$ (Matched)	59.0%	90.4%	59.6%	101.7

Comparatively, the absolute best performance achieved by the tuned collision classifier across 35 grid combinations was 63.2% (at $\lambda_c = 10.0$). The CBF explicitly outperforms the optimized classifier by 12.4 percentage points under mathematically identical environments. Furthermore, achieving 63.2% with the classifier required massive scalar weighting ($\lambda_c = 10.0$) that severely suppressed goal-seeking behavior (dropping Goal Rate to 79.2%). The CBF avoids this gradient conflict entirely via its modular predictor-corrector formulation.

6.2 Multi-Position Generalization Results

When conditioning the barrier on obstacle coordinates (x, y) , counter-intuitive but highly promising results emerge.

Table 6.3: Key α Ablation Results for MultiPosBarrierNet

α	Success Rate	Goal Rate	CF Rate
0.10 (Best)	83.8%	93.6%	89.8%

The optimal performance is found at $\alpha = 0.10$, achieving an 83.8% Success Rate (93.6% Goal, 89.8% CF). The multi-position network actually outperforms the specialized fixed network. This indicates that forcing the neural network to encode a varied 2D spatial manifold provides a richer, more regularized gradient field than learning a single, static trajectory map.

A breakdown by position reveals that obstacle $[0.5, 0.0]$ remains the most challenging (73.6% success), directly intersecting the natural goal approach vector, while obstacle $[0.6, 0.2]$ allows a 93.6% success rate, validating the network’s spatial awareness.

Table 6.4: Per-Position Breakdown at Optimal $\alpha = 0.10$

Obstacle Position (x, y)	Success Rate	Goal Rate	CF Rate
$(0.4, -0.2)$	81.6%	98.4%	83.2%
$(0.5, 0.0)$	73.6%	86.4%	86.4%
$(0.6, 0.2)$	93.6%	98.4%	95.2%
$(0.5, 0.3)$	86.4%	91.2%	94.4%

6.3 General Obstacle Planning Results

While the simplified variants unequivocally validate the CBF formulation, the fully general BarrierNet $B(z, o)$ exhibits significant performance degradation.

Table 6.5: Key α Ablation Results for General BarrierNet

α	Success Rate	Goal Rate	CF Rate
0.00	19.2%	40.4%	58.4%
0.05 (Best)	33.0%	69.4%	39.8%
0.10 – 1.00	$\sim 28.0\% - 31.0\%$	up to 75.8%	$\sim 30.0\%$

The optimal setting ($\alpha = 0.05$) yields only a 33.0% Success Rate. The collapse in performance is attributed to two critical factors:

- **Curse of Dimensionality:** Expanding the input space to 11 dimensions requires mapping a highly complex topological boundary for random radii and heights. The 15,000 transition training scenarios proved insufficient for this generalization.
- **Distribution Mismatch (Covariate Shift):** The most devastating factor is identified through the metric `avg_min_B`. During planning, the average pre-correction barrier value was -0.2786 . Because the training transition dataset was generated using an obstacle-blind nominal planner, the barrier network routinely encounters states that are deeply unsafe ($B < 0$). If $B(z_k)$ is negative, the target $(1 - \alpha\Delta t)B(z_k)$ is also negative, demanding no meaningful constraint enforcement. The CBF is mathematically designed to maintain forward invariance of a safe set, not to recover from deep constraint violations.

6.4 Computational Analysis

Despite adding an autograd calculation for ∇B , the CBF formulation is significantly faster than the baseline. Average planning time for the baseline classifier was 186.3 ms. The fixed CBF

($\alpha = 0.20$) executed in 94.8 ms. This speedup occurs because the classifier calculates and back-propagates a collision penalty at every step, whereas the CBF incurs the autograd penalty only when the constraint is actively violated (approximately 37% of the time).

Chapter 7

Conclusion and Future Work

7.1 Summary of Contributions

This thesis successfully introduced and validated a neural Control Barrier Function framework operating directly within the latent space of a generative robotic planner. By decoupling the obstacle avoidance constraint from the goal-seeking objective and relying on a first-order Taylor expansion correction, the proposed model eradicated the cliff-edge gradient problems inherent to BCE classifiers.

Empirical validation was unequivocal on constrained tasks: the CBF mechanism achieved 75.6% and 83.8% success rates on fixed and multi-position tasks respectively, vastly outperforming optimized classifier benchmarks and executing nearly twice as fast. A rigorous characterization of the α parameter successfully mapped the delicate equilibrium between forward invariance guarantees and functional goal pursuit.

7.2 Limitations

The primary limitation exposed in this work is the catastrophic covariate shift present in the generalized random-obstacle planning task. A barrier function trained on naive, uncorrected trajectories cannot effectively enforce invariance when deployed in an interactive optimization loop that inevitably strays into unmapped topological regions of the 11-dimensional space. Furthermore, the first-order approximation assumption holds cleanly for linear dynamics, but the highly non-linear nature of the VAE decoder means that safe linear steps in z -space may occasionally unfold into warped, unsafe maneuvers in physical configuration space.

7.3 Future Work

The findings of this research lay immediate groundwork for three advanced formulations:

- **Dagger-style Iterative Training:** To resolve the covariate shift, transition data must be recursively generated using the CBF-corrected planner itself, feeding deployment distributions back into the training loop (Ross et al., 2011).
- **Latent Model Predictive Control (MPC):** The current system generates complete open-loop trajectories. Transitioning to a closed-loop Model Predictive Control framework operating sequentially within the latent space will allow the CBF to react to execution-time decoding errors.

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Appendix A

Implementation and Hyperparameter Details

This appendix details the exact layer configurations, dimensions, and activation functions for the neural networks implemented in this thesis. All networks were implemented using PyTorch.

A.1 A.1 Variational Autoencoder (VAE)

The VAE learns the joint distribution of configurations $q \in \mathbb{R}^7$ and end-effector positions $e \in \mathbb{R}^3$. The input state is $x = (q, e) \in \mathbb{R}^{10}$.

Encoder Network:

- Input Layer: $\mathbb{R}^{10} \rightarrow \mathbb{R}^{2048}$, ELU activation
- Hidden Layers: 3 layers of $\mathbb{R}^{2048} \rightarrow \mathbb{R}^{2048}$, ELU activation
- Latent Mean (μ): Linear mapping $\mathbb{R}^{2048} \rightarrow \mathbb{R}^7$
- Latent Log-Variance ($\log \sigma^2$): Linear mapping $\mathbb{R}^{2048} \rightarrow \mathbb{R}^7$

Numerical Stability Note: The variance is strictly bounded using the Softplus function: $\sigma^2 = \text{softplus}(\text{fc}_{22}(h)) + 1e^{-5}$.

Decoder Network:

- Input Layer: $z \in \mathbb{R}^7 \rightarrow \mathbb{R}^{2048}$, ELU activation
- Hidden Layers: 3 layers of $\mathbb{R}^{2048} \rightarrow \mathbb{R}^{2048}$, ELU activation
- Output Layer: Linear mapping $\mathbb{R}^{2048} \rightarrow \mathbb{R}^{10}$ (No final activation; outputs are in normalized space).

A.2 A.2 Collision Classifier (VAEObstacleBCE)

The collision classifier shares and freezes the VAE encoder weights. The classifier head takes the concatenated 7D sampled latent vector z and the 4D normalized obstacle parameters $o = [x, y, h, r]$.

Classifier Head:

- Input Concatenation: $[z, o] \in \mathbb{R}^{11}$
- Initial Projection (fc32): $\mathbb{R}^{11} \rightarrow \mathbb{R}^{2048}$, ELU activation
- Hidden Blocks (fc_obs): 3 layers of $\mathbb{R}^{2048} \rightarrow \mathbb{R}^{2048}$.
- *Implementation Specifics*: Unlike the VAE, the classifier applies a pre-activation ELU pattern: $h_{l+1} = W_l(\text{ELU}(h_l)) + b_l$.
- Output Layer (fc42): Linear mapping $\mathbb{R}^{2048} \rightarrow \mathbb{R}^1$ (Logit)
- Activation: Sigmoid $\sigma(\cdot)$ applied to logit to yield $P(\text{collision})$.

A.3 A.3 Control Barrier Function Networks

The Barrier networks abandon the Sigmoid activation to predict an unbounded scalar $B \in (-\infty, \infty)$.

General BarrierNet $B(z, o)$:

- Input: $[z \in \mathbb{R}^7, o \in \mathbb{R}^4] \rightarrow \mathbb{R}^{11}$
- Hidden Layers: 4 layers of $\mathbb{R}^{2048} \rightarrow \mathbb{R}^{2048}$, ELU activation
- Output: Linear mapping $\mathbb{R}^{2048} \rightarrow \mathbb{R}^1$

MultiPosBarrierNet $B(z, x, y)$:

- Input: $[z \in \mathbb{R}^7, \text{pos} \in \mathbb{R}^2] \rightarrow \mathbb{R}^9$
- Hidden Layers: 4 layers of $\mathbb{R}^{2048} \rightarrow \mathbb{R}^{2048}$, ELU activation
- Output: Linear mapping $\mathbb{R}^{2048} \rightarrow \mathbb{R}^1$

FixedBarrierNet $B(z)$:

- Input: $z \in \mathbb{R}^7$

- Hidden Layers: 4 layers of $\mathbb{R}^{2048} \rightarrow \mathbb{R}^{2048}$, ELU activation
- Output: Linear mapping $\mathbb{R}^{2048} \rightarrow \mathbb{R}^1$

Appendix B

Hyperparameter Configurations

This appendix catalogs the explicit hyperparameters utilized for model training and latent-space trajectory optimization, ensuring full reproducibility.

B.1 B.1 Training Configurations

Table B.1: Model Training Hyperparameters

Parameter	VAE	Classifier	BarrierNets (All)
Optimizer	Adam	Adam	Adam
Learning Rate (LR)	$1e^{-4}$	$1e^{-4}$	$1e^{-4}$
LR Scheduler	None	None	Cosine Annealing (to $0.01 \times \text{LR}$)
Batch Size	4096	4096	4096 (Split: Label / Transition)
Max Epochs	16,000	16,000	Variable (Early stopping based on metric)
Gradient Clipping	None	None	1.0 (L2 Norm)
GECO Parameters			
g_{goal} (MSE threshold)	0.0008	N/A	N/A
g_{lr}	0.005	N/A	N/A
g_{α}	0.95	N/A	N/A
CBF Loss Parameters			
Safe Margin γ	N/A	N/A	1.0
Loss Weights $(\lambda_s, \lambda_u, \lambda_d)$	N/A	N/A	(1.0, 1.0, 1.0)

B.2 B.2 Planning and Inference Configurations

The hyperparameter configurations for Activation Maximization differ between the original baseline and the isolated ablation studies.

Table B.2: Activation Maximization Planning Hyperparameters

Parameter	Reported Baseline	Classifier Ablation	CBF Ablation
Optimizer	Adam	Adam	Adam
Learning Rate	0.15	0.03	0.03
Max Optimization Steps	300	300	300
Success Threshold	10 mm	10 mm	10 mm
Prior Weight λ_{prior}	0.7	$\{0.001 - 0.1\}$	0.01
Collision Weight $\lambda_{collision}$	0.5	$\{0.1 - 10.0\}$	N/A
Classifier Temperature T	3.0	1.0	N/A
Planning GECO			
Enabled	True	False	False
α_{GECO}	0.008	N/A	N/A
τ_{prior}, τ_{obs}	6.0, 2.0	N/A	N/A
CBF Correction			
Decay Rate α	N/A	N/A	Swept $\{0.00 - 1.00\}$
Time Step Δt	N/A	N/A	1.0
Max λ_{cbf} limit	N/A	N/A	1.0
Adam State Reset	N/A	N/A	True (Triggered if $\lambda_{cbf} > 0$)

Appendix C

Dataset and Environment Properties

C.1 C.1 Free-Space and Collision Datasets

The training regime relies on offline datasets mapped to the continuous joint limits of the Franka Emika Panda manipulator.

- **RobotStateDataset (Free-Space):** Contains 100,000 samples of physically valid $(q \in \mathbb{R}^7, e \in \mathbb{R}^3)$ pairs. Features are z-score normalized. If the standard deviation of any dimension $\sigma < 1e^{-8}$, it is clamped to 1.0 to prevent division by zero.
- **RobotObstacleDataset (Collision):** Contains 100,000 samples parameterized as $(q, e, obs, label)$.
- *Cross-Dataset Normalization Constraint:* To ensure spatial consistency, the (q, e) subsets of the Collision dataset are explicitly normalized utilizing the statistical mean and variance metrics extracted from the Free-Space dataset. Binary labels $c \in \{0, 1\}$ bypass normalization.

C.2 C.2 CBF State and Transition Datasets

Training the BarrierNet requires capturing both the spatial boundaries and the temporal optimization dynamics.

- **CBFStateLabelDataset:** Generated by passing 15,000 collision scenarios deterministically through the frozen VAE encoder (using the mean μ , bypassing reparameterization noise) to yield $(z, o, label)$ tuples.
- **CBFTransitionDataset:** Generated by running a *nominal, obstacle-blind* planner on 15,000 scenarios. Waypoints are collected at each step yielding the tuple: $(z_k, z_{nom}, o, label_k, label_{nom})$.

- *Data Imbalance:* Because the nominal planner ignores obstacles, the resulting transition dataset inherently contains a significant fraction of mathematically unsafe transitions ($B_k < 0$), fundamentally influencing the network’s capacity to maintain strict forward invariance versus performing recovery projections.

C.3 C.3 Simulation and Evaluation Geometry

The independent geometric validation pipeline utilizes `Robo3D`, simulating the Franka Panda using capsule primitives.

- **End-Effector Offset:** The target planning coordinate includes a physical offset of $[0, 0, 0.1654]$ m along the Z -axis of the final frame, representing the tool center point (TCP).
- **Unit Conversions:** The VAE latent planner operates entirely in radians. Waypoints are dynamically converted to degrees prior to geometric validation within the ‘Robo3D’ capsule-intersection routines.